

MTAPI Hosting Guide

MTAPI Hosting Guide

Date: 2026-09-16 · Status: **Decision made** · Owner: TBD

This document covers everything related to hosting the MTAPl Docker bridge: infrastructure options, cost comparisons, the chosen setup (Contabo + nginx), and how the worker connects to it.

1. What are we hosting?

MTAPI is a REST API bridge that connects to MetaTrader 5 (and 4) broker servers. It runs as a Docker container (timurila/mt5rest or mtapiio/mt5rest, ~92MB). The container includes the MT5 terminal + a .NET REST wrapper.

We host this bridge ourselves. The worker (on Railway) sends HTTP requests to the bridge, which forwards them to the broker over the MT5 protocol.

Worker (Railway) → HTTP/HTTPS → MTAPl Bridge (Contabo) → MT5 protocol → Broker (Exness)

What the Docker container actually is

The Docker container is software that does two things:

1. **Runs a MetaTrader 5 terminal** — the same trading platform that runs on a desktop, but running on a server inside the container.
2. **Exposes a REST API** — an HTTP interface so our code can talk to it.

When our worker needs to place a trade, it sends an HTTP request to the container. The container receives that request, converts it into the MT5 protocol that the broker understands, and sends it to the broker's server. The broker responds, and the container sends the result back as JSON.

The container holds the broker connection open. Our worker does not connect to the broker directly — it connects to the container, and the container connects to the broker.

FXSocket does the same thing, but they run the terminals on their servers. With the Docker container, we run the terminals on our own server. That is the only difference.

2. Hosting options compared

2.1 MTAPI Cloud (hosted by mtapi.io)

Tier	Price	Accounts	Notes
Trial	Free	1	14-day trial, token expires ~5min inactivity
Cloud 100	\$500/mo	100	Managed hosting, mtapi.io runs the bridge
Cloud 500	\$1,500/mo	500	Same, more capacity
On-Premise	\$1,000/mo	Unlimited	License for self-hosted Docker

Pros: No ops burden. Managed infrastructure. TLS included. **Cons:** Expensive. \$500/mo minimum for production. Token-based auth only (no API key).

2.2 Railway (where the worker already runs)

Component	Cost
MTAPI bridge (2 vCPU, 4 GB RAM)	~\$80-100/mo
Worker (existing)	~\$50/mo
Total	~\$130-150/mo

Railway pricing: \$20/vCPU/month + \$10/GB RAM/month + plan fee (\$5-20/mo).

Pros: Same platform as worker. Easy to manage. Auto-restarts. **Cons:** Usage-based billing = expensive for an always-on service. Overkill for a simple Docker container. Egress charges (\$0.05/GB).

2.3 Hetzner

Plan	CPU	RAM	Storage	Traffic	Price (US)	Price (EU)
CPX11 (shared)	2 vCPU	4 GB	80 GB NVMe	1 TB	\$20.49/mo	€5.99/mo
CPX21 (shared)	4 vCPU	8 GB	160 GB NVMe	1 TB	\$37.49/mo	€9.99/mo
CPX31 (shared)	8 vCPU	16 GB	320 GB NVMe	1 TB	\$73.49/mo	€17.99/mo
CCX13 (dedicated)	2 vCPU	8 GB	80 GB NVMe	1 TB	\$50.99/mo	€16.49/mo

Note: Prices increased ~30-40% on June 15, 2026. US datacenters (Ashburn VA, Hillsboro OR) only offer CPX/CCX — no CX/CAX budget tiers. EU regions (Germany, Finland) are 3-4x cheaper but add ~80ms latency to NY broker servers. US plans include only 1 TB traffic (vs 20 TB in EU); overage at \$1/TB.

Pros: Excellent performance (Geekbench 6 single-core ~1,480, NVMe 58k IOPS). Ashburn VA is the world's densest internet exchange — great peering. Provisioning in ~25 seconds. 99.9% SLA. Clean API and Terraform provider. **Cons:** US pricing is 3-4x EU pricing. Only 1 TB traffic included in US. No phone support (ticket + email only). Servers bill whether on or off.

2.4 Contabo

Contabo has three product tiers. The difference is how much of the physical CPU you actually get:

Tier	CPU model	Starting price	Notes
Cloud VPS	Shared vCPUs (time-sliced with other tenants)	\$5.28/mo	What we use. Cheap, good enough for I/O-bound work.
Cloud VDS	Dedicated cores (no sharing)	~\$49/mo	Consistent performance, no noisy neighbours. Consider if we see latency variance.
Dedicated Server	Entire bare-metal machine	~\$149/mo	Overkill for our workload.

Our plan — Cloud VPS (Core), US pricing, 24-month term:

Plan	CPU	RAM	Storage	Traffic	Base price
Cloud VPS 4	4 vCPU	8 GB	100 GB SSD	Unlimited	\$5.28/mo
Cloud VPS 6	6 vCPU	12 GB	200 GB SSD	Unlimited	\$7.20/mo
Cloud VPS 8	8 vCPU	24 GB	300 GB SSD	Unlimited	\$13.44/mo
Cloud VPS 12	12 vCPU	48 GB	400 GB SSD	Unlimited	\$24.00/mo
Cloud VPS 16	16 vCPU	64 GB	500 GB SSD	Unlimited	\$35.60/mo
Cloud VPS 18	18 vCPU	96 GB	600 GB SSD	Unlimited	\$47.04/mo

US datacenter locations: **US East (Carlstadt, NJ)**, US Central (St. Louis), US West (Seattle). Location fees apply on top of the base price:

Region	Location fee
EU (Germany)	Free
US Central (St. Louis)	\$1.80/mo
US West (Seattle)	\$2.30/mo
US East (New York)	\$2.80/mo

Actual cost for US East (our use case): \$5.28 + \$2.80 = **\$8.08/mo** for Cloud VPS 4 (24-month term).

Plans can be upgraded in place from the Contabo control panel — RAM, vCPU, and storage can be scaled without rebuilding the server. So we do not need to over-provision up front; start on VPS 4 and upgrade when monitoring says so.

Pros: Cheapest option — 4 vCPU + 8 GB for \$8.08/mo in NY. Unlimited traffic. Carlstadt NJ is ~5ms from NYC, ~15ms from NY broker servers. 99.9% SLA. **Cons:** Shared vCPUs with lower per-core performance. Email-only support. Provisioning can take hours on first order. No hourly billing.

Pricing note: Contabo shows different figures depending on locale (USD vs EUR) and whether VAT is included. The USD figures above are from the US site (contabo.com/en-us/vps) as of 2026-09-17. Confirm at order time — Contabo changes prices without notice.

2.5 Other VPS providers (for reference)

Provider	4 vCPU, 8 GB RAM	Notes
DigitalOcean	~\$24-40/mo	Better docs, marketplace
Vultr	~\$20-40/mo	Good global coverage
AWS Lightsail	~\$20-40/mo	Easy AWS integration

2.6 Broker landscape

User accounts by broker (production, as of 2026-09-16):

Broker	Accounts	Server names	Primary datacenter
Exness	31		NY (us-east-1)

Broker	Accounts	Server names	Primary datacenter
		MT5Trial9, MT5Trial15, MT5Trial11, MT5Real10, ...	
VantageMarkets	27	Demo, Live 14, Live 4, Live, Live 6, ...	LD4 (London) + NY
PUPrime	14	Demo, Live 6, Live	LD4 (London) + HK
ICMarkets	12	SC-Demo, SC- MT5-3, SC- MT5-4, SC- MT5-2	NY (us-east-1) + LD4
MetaQuotes	7	Demo (MTAPI demo server)	Global
Tickmill	6	EU-Demo, Demo	LD4 (London)
XMGlobal	5	Demo 2, MT5 2/4/7/9	LD4 + NY
FTMO	5	Demo, Demo2	LD4 (London)
RoboForex	4	Pro, ProCent-5	LD4 + HK
4xHub	4	International- Server	LD4 (London)
VEOMarkets	3	Trade	LD4 (London)
FBS	2	Demo	LD4 + NY
BlackBull	2	Demo, Live	LD4 (London)
Pepperstone	1	UK-Demo	LD4 (London)
Other	53	(40+ different brokers)	Various

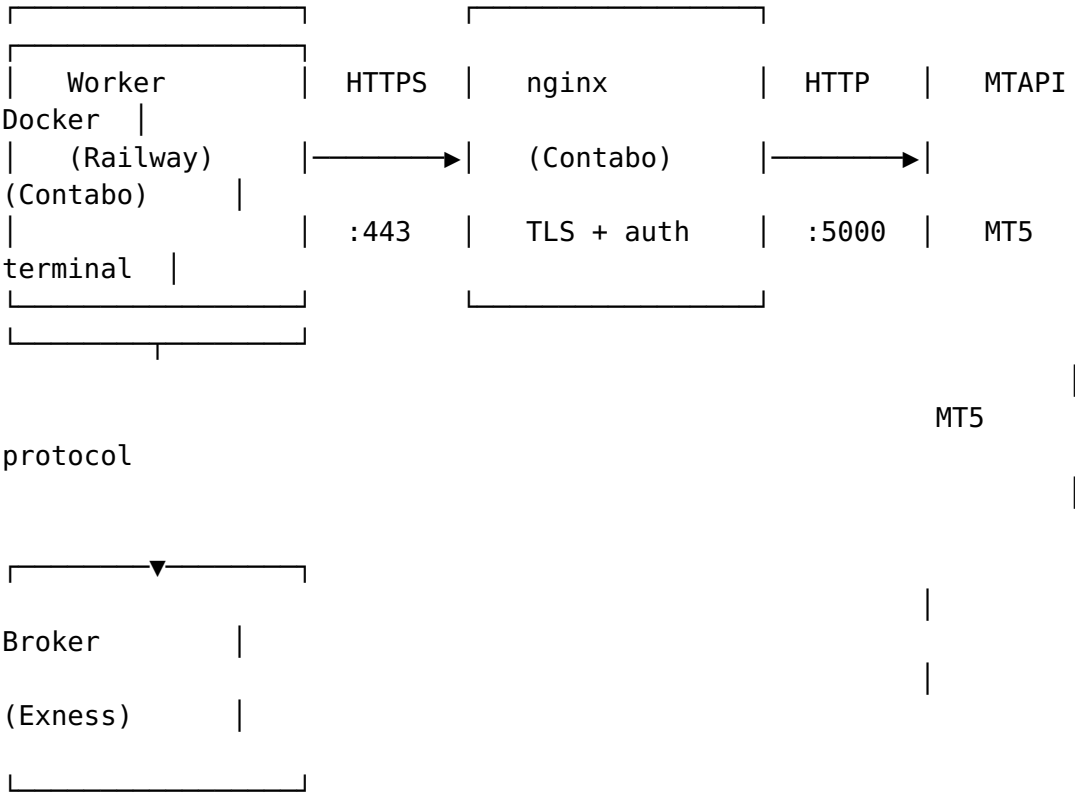
Top 4 brokers = 84 accounts (60% of userbase). All have NY or LD4 servers. Hetzner NY gives the lowest latency to the majority of broker endpoints.

3. Cost summary

Scenario	Monthly cost
MTAPI Cloud 500	\$1,500
MTAPI Cloud 100	\$500
Railway (MTAPI + Worker)	~\$130-150
Contabo Core VPS 4 (US East) + Worker (Railway)	~\$58
Hetzner CPX21 (US) + Worker (Railway)	~\$87
Contabo Core VPS 4 (US East) only	~\$8
Hetzner CPX21 (US) only	~\$37

Decision: Contabo Core VPS 4 in US East (Carlstadt, NJ) for the MTAPI bridge. At \$8.08/mo (\$5.28 base + \$2.80 location fee) with 4 vCPU, 8 GB RAM, and unlimited traffic, it delivers the MTAPI bridge workload at one-quarter the cost of Hetzner US. Carlstadt NJ is ~5ms from NYC, ~15ms from NY broker servers. Worker stays on Railway.

4. Architecture



Why nginx?

1. **TLS encryption** — encrypts traffic between Railway and Contabo
2. **Authentication** — shared secret header (X-MTAPI-Key) proves identity
3. **Rate limiting** — can throttle abusive requests
4. **Logging** — logs all requests for debugging

MTAPI itself has no API key. Security is network-level (nginx + firewall).

5. Setup guide

5.1 Contabo VPS

1. Sign up at contabo.com
2. Create a Cloud VPS 4 (Ubuntu 22.04) — select **US East (New York)** datacenter
3. Note the IP address and root password

5.2 Initial server setup

```
# SSH into the server
ssh root@<contabo-ip>

# Update system
apt update && apt upgrade -y

# Install Docker
curl -fsSL https://get.docker.com | sh
systemctl enable docker

# Install nginx
apt install -y nginx

# Install certbot (for TLS)
apt install -y certbot python3-certbot-nginx
```

5.3 Deploy MTAPI Docker

```
docker run -d \
  --name mt5rest \
  --restart always \
  -p 5000:80 \
  timurila/mt5rest
```

Verify:

```
docker logs mt5rest
curl http://localhost:5000/AccountSummary?id=test-token
```

5.4 nginx configuration

Create /etc/nginx/sites-available/mtapi:

```
server {
    listen 443 ssl;
    server_name mtapi.yourdomain.com;

    ssl_certificate /etc/letsencrypt/live/mtapi.yourdomain.com/
fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/
mtapi.yourdomain.com/privkey.pem;

    # Shared secret auth
    location / {
        if ($http_x_mtaapi_key != "YOUR_SECRET_KEY_HERE") {
            return 403;
        }

        proxy_pass http://127.0.0.1:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For
$proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;

        # Timeouts for long-running requests
        proxy_connect_timeout 30s;
        proxy_read_timeout 60s;
        proxy_send_timeout 60s;
    }
}
```

Enable and get TLS cert:

```
ln -s /etc/nginx/sites-available/mtapi /etc/nginx/sites-enabled/
nginx -t
systemctl reload nginx

# Get TLS certificate
certbot --nginx -d mtapi.yourdomain.com
```

5.5 Firewall

```
# Allow SSH
ufw allow 22/tcp

# Allow HTTPS (nginx)
ufw allow 443/tcp

# Allow HTTP (certbot renewal)
ufw allow 80/tcp

# Enable firewall
ufw enable
```

Port 5000 (MTAPI Docker) is NOT exposed publicly — nginx proxies to it locally.

5.6 DNS

Point a subdomain to the Contabo IP:

mtapi.yourdomain.com → <contabo-ip>

6. Worker connection

6.1 Environment variables (Railway)

```
MTAPI_BASE_URL=https://mtapi.yourdomain.com
MTAPI_KEY=YOUR_SECRET_KEY_HERE
```

6.2 Worker code

Every MTAPl request includes the auth header:

```
const response = await fetch(`${MTAPI_BASE_URL}/AccountSummary?
id=${token}`, {
  headers: {
    'X-MTAPl-Key': process.env.MTAPl_KEY,
  },
});
```

6.3 Request flow

1. Worker sends: GET https://mtapi.yourdomain.com/AccountSummary?
id=<token>

Header: X-MTAPI-Key: <secret>

2. nginx receives request
 - Validates X-MTAPI-Key header → 403 if wrong
 - Forwards to `http://127.0.0.1:5000/AccountSummary?id=<token>`
 3. MTAPI Docker receives request
 - Looks up session by token
 - Queries broker via MT5 protocol
 - Returns JSON response
 4. nginx returns response to worker
-

7. Security model

Layer	What it protects
nginx X-MTAPI-Key	Proves request came from our worker (not public internet)
TLS (Let's Encrypt)	Encrypts traffic between Railway and Contabo
Contabo firewall	Blocks all ports except 22, 80, 443
MTAPI token (?id=)	Identifies the broker session (not authentication)
Broker credentials	MTAPI stores login/password to connect to Exness

MTAPI has no API key. The X-MTAPI-Key is an nginx-level shared secret. MTAPI never sees this header — nginx strips it before forwarding.

8. Monitoring

8.1 Health check

```
# From worker or anywhere with access
curl -H "X-MTAPI-Key: <secret>" https://mtapi.yourdomain.com/
CheckConnect?id=<token>
# Expected: OK
```

8.2 Docker logs

```
ssh root@<contabo-ip>  
docker logs -f mt5rest
```

8.3 nginx logs

```
tail -f /var/log/nginx/access.log  
tail -f /var/log/nginx/error.log
```

8.4 Uptime monitoring

Set up a cron job or external monitor to ping:

```
GET https://mtapi.yourdomain.com/CheckConnect?id=<token>  
Header: X-MTAPI-Key: <secret>  
Expected: 200 OK
```

8.5 Capacity and upgrade triggers

There are two independent upgrades: the **Docker image** (software) and the **VPS plan** (hardware). Do not confuse them.

Docker image upgrades

Trigger	How to detect
Trial expiry	The trial image works for 14 days, then stops accepting new connections.
Security patches	Run <code>docker scout cves timurila/mt5rest</code> monthly. CVE count drops when MTAPI publishes an updated image.
New API features	MTAPI announces it, or we need an endpoint the running image lacks.

Check for a newer image and re-scan:

```
docker pull timurila/mt5rest  
docker scout cves timurila/mt5rest
```

To upgrade: stop and remove the old container, pull the new image, and run it with the same flags. Sessions reconnect automatically via stored credentials.

VPS plan upgrades

The container exports a Prometheus gauge `myapp_running_count_mt5` — the current number of active broker sessions. Alert when it approaches `MaxSessions`.

Measured memory footprint (2026-09-17, trial image, Exness demo):

State	Memory
Container idle	55.9 MB
1 session connected	88–91 MB
Marginal cost per session	~35 MB

Memory was stable — it did not grow over time or after API calls. Based on this:

Sessions	Estimated RAM
50	~1.8 GB
163	~5.6 GB
300	~10.5 GB
500	~17.5 GB

Signal	Meaning	Action
Sessions near <code>MaxSessions</code>	Server is filling up	Add a second VPS or upgrade the plan
RAM consistently > 80%	More sessions than RAM allows	Upgrade to Cloud VPS 6 (12 GB) or 8 (24 GB)
CPU consistently > 80%	Too many sessions per core	Upgrade or scale out
Trade latency rising	Server is overloaded or network degraded	Investigate immediately (see §8.6)

Capacity rule of thumb: Cloud VPS 4 (4 vCPU, 8 GB) should carry **~163 sessions** at ~5.6 GB, leaving ~2 GB for the OS and nginx. Upgrade to Cloud VPS 6 (12 GB, \$10/mo) for comfort or to target 300+ sessions. Plans upgrade in place — no rebuild needed.

Caveat: The per-session figure was measured with one session on one broker. Other brokers, or sessions that subscribe to many quotes, may use more. Verify with 3–5 demo accounts before finalizing the production plan.

A lightweight cron that scrapes `/metrics` and alerts when sessions > 100 or RAM > 80% is sufficient for now. A full Prometheus/Grafana stack is not required yet.

8.6 Measuring latency

Trading is latency-sensitive, so we measure at three layers:

Layer	What it measures	How
Worker → VPS	Network round-trip (Railway to Contabo)	Synthetic probe: call <code>AccountSummary</code> or <code>/healthz</code> every 60s and record duration
VPS → broker	MTAPI's internal broker round-trip	nginx <code>\$upstream_response_time</code> in the access log
End-to-end	Signal dispatch → broker confirmation	Duration logged per <code>requestWithRetry</code> call in <code>mtapiProvider.ts</code>

1. Instrument the provider (best signal). Add duration logging to `requestWithRetry` in `worker/src/mtapiProvider.ts` — record endpoint, duration in ms, and success/failure. Aggregate p50/p95/p99 per endpoint. This measures real trade latency, not synthetic probes.

2. nginx timing. Add to the nginx log format:

```
log_format timing '$remote_addr - $request_time
$upstream_response_time "$request"';
access_log /var/log/nginx/access.log timing;
```

`$upstream_response_time` is how long the MTAPl container took;
`$request_time` includes nginx overhead.

3. Synthetic probe. A cron on the worker (or an external monitor) calls a cheap endpoint every 60s and records the round-trip. This catches network degradation between Railway and Contabo before it affects trades.

4. Compare against FXSocket. Both providers run in parallel on different accounts. `trade_execution_logs` already records broker timing — compare MTAPI vs FXSocket execution times directly.

Suggested alert thresholds:

Metric	Warning	Critical
Trade latency p95	> 500 ms	> 1000 ms
Trade latency p99	> 1000 ms	> 2000 ms
Synthetic probe p95	> 200 ms	> 500 ms
CPU sustained	> 80%	> 90%
Memory	> 80%	> 90%

Why this matters: Shared vCPUs on Cloud VPS can cause latency variance when a neighbour spikes. If p95 latency becomes unstable, that is the signal to move to the Cloud VDS tier (dedicated cores).

9. Backup and recovery

9.1 What to back up

- nginx config: `/etc/nginx/sites-available/mtapi`
- TLS certs: `/etc/letsencrypt/live/mtapi.yourdomain.com/`
- Docker container is stateless — no data to back up

9.2 Recovery

If the VPS dies: 1. Create new Contabo VPS 2. Run setup script (§5.2) 3. Deploy Docker container (§5.3) 4. Copy nginx config + TLS certs 5. Update DNS if IP changed

The MTAPI bridge is stateless — sessions are stored in-memory and re-created on connect. The worker handles reconnection via `ConnectByToken`.

10. Scaling

Current sizing (Cloud VPS 4): - 4 vCPU, 8 GB RAM - Handles ~150 concurrent MTAPI sessions (each session = one broker account) - MTAPI Docker uses ~200-500MB RAM per 100 sessions

If we exceed 150 accounts: - Upgrade to Cloud VPS 6 (\$7.95/mo) for 6 vCPU, 12 GB RAM - Or Cloud VPS 8 (\$15/mo) for 8 vCPU, 24 GB RAM

11. Rollback

If MTAPI hosting fails: 1. Set all accounts back to provider = 'fxsocket' in the database 2. Worker automatically routes to FXSocket (no deploy needed) 3. Shut down Contabo VPS (cancel or stop) 4. FXSocket is a hosted SaaS — it's always available

12. Follow-ups

1. Buy a domain or subdomain for the MTAPI bridge (~\$1/year)
2. Set up the Contabo VPS (\$5.2-5.5)
3. Deploy MTAPI Docker and nginx
4. Configure Railway env vars (MTAPI_BASE_URL, MTAPI_KEY)
5. Test connectivity from worker
6. Set up uptime monitoring